

Toumi Adem

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SUMMARY

AI & Data Science Engineer trained at École Supérieure en Informatique de Sidi Bel Abbès (ESI-SBA), specializing in artificial intelligence, machine learning, data science, and software engineering. Experienced in log anomaly detection, computer vision, natural language processing, reproducible experimentation, and reliable systems.

EDUCATION

École Supérieure en Informatique de Sidi Bel Abbès (ESI-SBA) <i>Engineering Degree & Master's in Artificial Intelligence and Data Science</i>	Sidi Bel Abbès, Algeria 2021 – 2026
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TECHNICAL SKILLS

AI & Data Science:	Python, Pandas, NumPy, Scikit-learn, TensorFlow, PyTorch, OpenCV, XGBoost, FastAPI
Data Engineering:	SQL, PostgreSQL, Apache Spark, Apache Kafka, MLflow
Programming:	Python, Java, Dart, JavaScript, TypeScript
Web & Mobile:	React, Next.js, Flutter, Node.js, HTML, Tailwind CSS, SQL, Android (Java)
Tools:	Git, Docker, Linux

EXPERIENCE

Research Intern – Object Detection & Tracking for UAV Systems <i>Electronic Systems Research and Development Unit (URD Laboratory), Sidi Bel Abbès, Algeria</i>	2024 – 2025
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- Prepared aerial object detection datasets, including VisDrone and military vehicle imagery, for YOLO training and benchmark evaluation.
- Trained YOLOv11s and evaluated on VisDrone validation set (mAP@0.5 $\approx$ 0.85); implemented DeepSort and CSRT tracking algorithms.
- Deployed edge inference on Raspberry Pi 5 with Intel Movidius NCS 2.

PROJECTS

Software Engineering

KidLink – School Communication Platform	<i>Academic Coursework, 2022 – 2023</i>
• Cross-platform Flutter mobile application for parent-teacher communication in primary schools; role-based access (parent, teacher, administrator) and real-time push notifications via Firebase Cloud Messaging. • Technologies: Dart, Flutter, Firebase (Auth, Firestore, FCM, Storage), Provider.	

HajjPortal – Hajj Pilgrimage Management Platform	<i>Academic Project, 2023 – 2024</i>
• Trilingual (Arabic, English, French) Next.js 14 web platform with full RTL/LTR layout switching; covers pilgrim registration, hotel reservations, lottery draws, and journey tracking. • Separate admin dashboard with Socket.io real-time updates and TanStack Query for server state management. • Technologies: TypeScript, Next.js 14, React, Tailwind CSS, Socket.io, Zod, React Hook Form.	

XAI Log Anomaly Detection – PFE Showcase	<i>Academic Showcase, 2024 – 2025</i>
• Single-page web showcase presenting the PFE experimental results across 9 interactive sections, including dataset overview, pipeline architecture, benchmark comparisons, and SHAP/LIME XAI gallery. • Technologies: React, Vite, Framer Motion, Recharts, Tailwind CSS.	

AI & Data Science

Log-Based Anomaly Detection in Distributed Systems *Engineering PFE & Master's Thesis, 2025–2026*

- Benchmarked 11 ML/DL models – SVM, Random Forest, Decision Tree, Isolation Forest, Attention-BiLSTM, CNN+BiLSTM, DeepLog, LSTM-AE, Dense-AE, LogBERT, BiLSTM-AE+Word2Vec – across HDFS, BGL, and Spirit log datasets under a unified experimental protocol.
- Conducted a systematic literature survey and built a real-time streaming pipeline: Apache Kafka → cloud-hosted HuggingFace Spaces → PostgreSQL → Streamlit dashboard with SHAP and LIME post-hoc explainability.
- **Technologies:** Python, PyTorch, TensorFlow, Scikit-learn, FastAPI, Apache Kafka, PostgreSQL, Streamlit, SHAP, LIME, Pandas.

Object Detection & Tracking for UAV Applications *Academic Internship, 2024–2025*

- Trained YOLOv11s on VisDrone and military vehicle datasets; integrated DeepSort and CSRT tracking algorithms ($\text{mAP}@0.5 \approx 0.85$).
- **Technologies:** Python, PyTorch, Ultralytics YOLO, OpenCV, DeepSort.

Paraphrase Detection with Siamese Neural Networks *NLP Project, 2024–2025*

- Siamese BiLSTM (3 layers, 512 hidden units per direction) with 4-head multi-head attention trained on PAWS and Quora Question Pairs (~ 450 K pairs); WordNet synonym augmentation and Focal Loss ($\alpha = 0.75$, $\gamma = 2$) for class imbalance.
- **Technologies:** Python, PyTorch, spaCy, NLTK.

Malicious URL Detection using Machine Learning *ML Project, 2024–2025*

- Compared Logistic Regression, Random Forest, XGBoost, and LSTM on 651,191 URLs across four classes (benign, defacement, phishing, malware); TF-IDF character n-grams (3–5 gram) combined with 15+ structural URL features; best result: Random Forest at 96.8% macro-averaged F1-score.
- **Technologies:** Python, Scikit-learn, XGBoost, TensorFlow, Pandas.

Movie Recommendation System *ML Project, 2024–2025*

- Implemented Apriori and FP-Growth association rule mining with MLflow experiment tracking across 60+ runs on MovieLens data.
- **Technologies:** Python, Scikit-learn, MLflow, Streamlit.

LANGUAGES

Arabic (Native) French (Fluent) English (Professional)